Unraveling Evolutionary Dynamics: Insights from *In Silico* Experiments on Selective Mechanisms in Controlled Environments

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Abstract

Driven by the purpose to address the intrinsic difficulties of studying the evolution of biological systems, in this paper, we present a series of *in silico* experiments aimed at exploring the evolutionary dynamics unfolding in a model of colorectal cancer development, using methods from evolutionary biology. The analysis span from the single gene level, to the phenotypic level. We begin with the standard model commonly used in population genetics to measure natural selection on allele frequencies. We then employ the Price equation, a well-established formalism known for its effectiveness in tracking evolving relationship between entities over time, such as differences between parental and offspring traits. The model results are finally analysed to explain how the selection mechanism works in the context provided by the simulated world. While natural selection is often considered as a statistical phenomenon resulting from changes in genotypic population frequencies, our work suggests that in models with extensive environmental control it is possible to identify the specific elements responsible for exerting selection pressure on the so-called selection units. In this way, we explore the possibility of deciphering the factors that drive and are driven by the selective process.

Keywords: Natural selection, Price equation, Colorectal cancer, Evolutionary theory, Computational biology

1. Introduction

Systems known as complex systems present a particular challenge. They comprise numerous entities from various hierarchical levels, engaging in intricate interactions that defy straightforward explanation through efficient causation. Usually, a linear or nonlinear deductive model is unable to grasp the past and future evolution of the aggregate behavior of heterogeneous entities. This complexity is particularly pronounced in biological systems, where entities span

different levels, with properties and dynamical relations shifting as one moves from one level to another [1]. Consider, for example, the unclear relationship between structure and function. In fields ranging from molecular to evolutionary biology, understanding how a particular function works in conjunction with its corresponding structure does not explain how the same structure can perform new functions in different environments or, conversely, how different structures can perform the same function [2, 3].

Conducting experiments in fields deeply rooted in history has always been a challenging task [4]. In classical experimental sciences like physics, chemistry, or molecular biology, scientists work with a limited number of variables in a controlled environment, assuming that both endogenous and exogenous perturbations of the system can be integrated into the analysis. Closed environments like labs or individual experiments allow for this level of control through the repeated testing of the same process while modifying one or more variables at a time, keeping the other features unchanged. On the other hand, when a historical dimension is inherent to the system, a significant challenge arises: it becomes impossible to intervene multiple times without altering the evolutionary history of the entity in question [5].

Moreover, biological systems often span extended time scales with varying rates of change, such as those observed in macroevolution and microevolution (even though the issue is debated [6]). Once the specific evolutionary history of a system has unfolded, it becomes impossible to restart the process by altering one or more variables in isolation to acquire causal insights. In fact, attempting to replay the historical process would yield different outcomes [7]. Nevertheless, a powerful way to comprehend biological systems is to reproduce them, or at least some of their characteristics, with computational methods, where relevant variables are already chosen, and environmental fluctuations are more easily manageable [8]. Agent-based models, a specific category of computational models, offer a valuable approach for examining such processes and addressing these issues. These models enable the compression of time scales and the selection of relevant variables, making them more manageable for researchers. Additionally, they facilitate the exploration of spatial dynamics, which are essential for understanding evolutionary changes in ecological and population-based contexts [9].

The aim of this paper is multifaceted: first, to utilize agent-based modeling to simulate the evolutionary dynamics of a specific biological system, i.e., the development of colorectal cancer (CRC) [10]; second, to observe how selection operates within the system, reasoning about its "causal power" along with unpacking the selective agents; and lastly, to investigate the evolutionary characteristics of the system and how they respond to interventions. The following sections will cover: a short description of the ABM; an analysis of the model using standard population genetics and Price's equation; a discussion about the evolutionary methods in cancer research; issues and open questions resulting from model findings.

2. Methods

2.1. Model's synthetic description

To model the development of CRC within the colonic crypt, we built an ABM reproducing the colonic crypt as an unfolded cylinder where 100 layers, of 25 cells each, perform several actions programmed at the individual level [10]. The key elements of the world are two physiological populations, divided into *transit-amplifying* and *differentiated* cells, occupying, respectively, the bottom part of the crypt and the top part, in a proportion of roughly 35% and 75%. This proportion, as well as the population size, is kept constant by three dynamical factors: the creation of a new bottom layer of cells at each time step; the exit from the simulation of all the cells reaching the top of the crypt; the decrease of β-catenin which starts the differentiation from *transit-amplifying* to *differentiated*.

Each cell has a genome formed by three driver genes (*APC*, *KRAS*, *TP53*), which control different functions: *APC* controls the differentiation of cells and movement; *KRAS* controls reproduction and movement; *TP53* controls reproduction and DNA integrity by means of augmenting the probability of mutation of the other genes when mutated. The three genes can mutate, according to the mutation probability, during the mitosis procedure, alone or jointly, either way affecting cellular behavior. Along with genetic features are environmental conditions: an immune-system cell population and a hypoxic threshold. Both conditions are triggered by a population threshold, that is, a given number of cells located in the same spot, which determines the activation of immune-system cells or the death of neoplastic cells.

All the possible genotypes produce four functional phenotypes upon which we concentrated our analysis, namely: *physiological*, *pre-adenoma*, *adenoma*, and *tumoral* cells[1]. The first one, *physiological*, belongs to cells that have all the alleles in wild-type status or in heterozygosis with no functional changes. In the model, genes have two copies represented as lists of binary values ([0/1, 0/1]), denoting the two alleles. These genes are associated with a sum threshold, which is linked to a cellular function such as movement or mitosis time. A value of 0 indicates the wild-type allele, while a value of 1 signifies a mutated allele. For instance, $(\sum_0^2 APC = 1)$ if either [1 0] or [0 1]. Cells with phenotypes corresponding to *physiological*, *pre-adenoma*, and *adenoma* have a maximum lifespan of 120 hours; *tumoral* cells have a maximum lifespan of 150 hours. In the following figure, the simulated crypt at different stages of tumor development is shown; see Fig. 1.

Runs have been performed for four significant scenarios based on mutation probability: $\mu = 10^{-9}, 10^{-6}, 10^{-4}, 10^{-2}$. Three stop conditions: 1 year of model simulation [11]; reversed proportion of *transit-amplifying* and *differentiated* cells [12]; population inside of the crypt equals to 10 thousand individuals [12].

[1] In the agent-based model, *typology* is the variable used to label the four analyzed phenotypes. In other words, we can say that phenotype = *typology*.

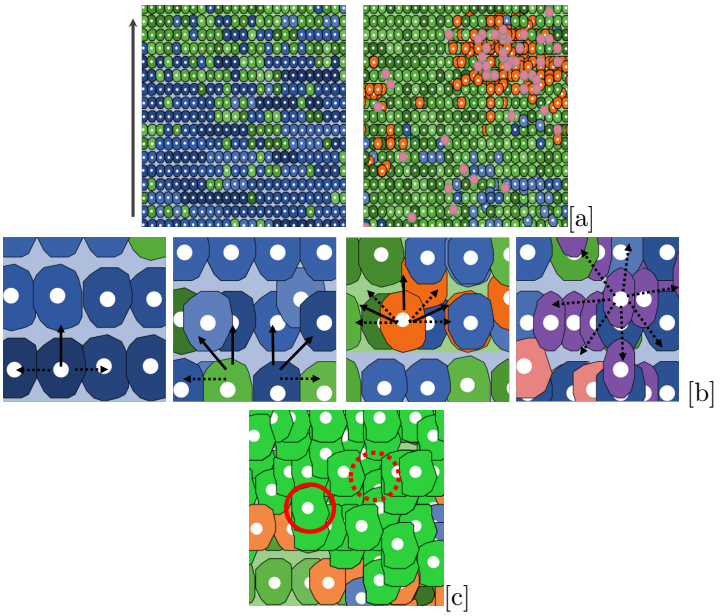


Figure 1: Figure [a] shows two local portions of the crypt. The left image depicts the physiological state, where *transit-amplifying* (blue) and *differentiated* (green) cells, move upward following the direction of the grey line. The picture on the right depicts a crypt where cells have higher probability of mutation, in fact are present cells with different colors (orange), along with immune-system cells (pink and grey). In [b] we observe, from the left to the right, the four typologies and their different movements and proliferation directions. Solid black lines represent some of the possible movement direction, while dashed black lines are the potential mitosis directions (where the daughter cell is going to appear). *Physiological* cells have 0° movement, i.e. forward movement; *pre-adenoma* and *adenoma* cells have a range spanning from 0° to 90°, both for movement and reproduction; *tumoral* cells reproduce with a 360° range. In [c] is shown a snapshot with neoplastic cells as to describe the hypoxic effect. The solid-line circle indicates the most superficial cell, and the dashed circle the inner one; when the hypoxic threshold is activated the inner cell is the first one to die and a cascading effect target the others until the value is again below the threshold.

2.1.1. Classic Population Genetics

Population genetics is the branch of evolutionary biology dedicated to tracking changes in gene frequencies across successive generations, incorporating fundamental mechanisms like natural selection, random genetic drift, mutation, and migration. These mechanisms exert their influence on biological populations either collectively or independently.

Even though there is evidence that neutral theory plays a considerable role in sub-clonal expansion and intratumoral heterogeneity [13, 14], for the purposes of this study, we will focus exclusively on natural selection, as it stands out as the most relevant mechanism when addressing adaptation to novel environmen-

tal conditions[2] [15]. Moreover, its importance as guiding principle of tumor progression has been recognized by several studies during the last decades, see for example Fortunato et al. for adaptation [16], Thomas et al. for selection for function [17], or Khong and Restifo presenting natural selection as the escaping mechanism of tumoral cells from immune system [18]. To quantify this variation, the appropriate formalism or mathematical framework is the fundamental model of natural selection [19, 20].

$$\Delta p = p_{t+1} - p_t = \Delta p = \frac{p(1-p)(w_p - w_q)}{\bar{w}} \tag{1}$$

When this quantity is represented as a function of the wild-type allele frequency p, it reveals the strength of variation for each allele frequency. If $\Delta p > 0$, natural selection has led to an increase in the frequency of A_1 in the population. Conversely, if $\Delta p < 0$, the selection process has caused A_1 to decrease in frequency. Finally, if $\Delta p = 0$, there is no change in the allelic frequency [21].

Although classical population genetics commonly assumes that populations exhibit sexual reproduction with random mating, we cannot assume such a condition. Indeed, this feature is absent in our model because the cells reproduce asexually, hence without actually mating with other cells.

Regarding the no-recombination assumption, which implies that only complete deletions or point mutations occur, this is also satisfied. The individuals (cells) undergo mitosis, producing clones without any genetic exchange with other cells. They are subject to random mutations, but the genetic material remains intact without recombination.

The only challenge arises from the assumption of non-overlapping generations, which our model does not fully satisfy. Cells do not die after their first mitotic division, so there is always a proportion of cells consisting not only of the contribution of p (or q) to the next generation but also of p (or q) itself.

2.1.2. Price formalism

In 1970, George R. Price made a significant contribution to evolutionary biology, and theoretical biology in general, by deriving the equation that bears his name [22, 23]. Price's equation aims to reveal correlations between essential properties (e.g., phenotypes and the number of offspring) at every level of the system under investigation. It serves as a mathematical identity that holds true irrespective of its applicability. Whenever there is a hypothesis aimed at explaining the evolution or change of a biological entity's trait and its associated fitness, this equation represents a valuable tool for uncovering such crucial relationships [24].

[2]For our purpose, it is not necessary to model *mutation* explicitly to observe, for instance, whether or not a mutation-selection equilibrium exists. The reason is that the mutation-selection model considers, in a simple two-allele scheme (A_1, A_2), the reappearance of an allele due to mutation $A_1 \longleftrightarrow A_2$. However, in our model the wild-type alleles return only from the activity of stem cells, not involving mutation from the neoplastic allele to the wild-type one within the crypt population.

Moreover, it has been demonstrated that the most abstract form of this equation is not limited to biological phenomena but extends to processes where information flows from one set to another, with the potential for replication errors [25]. Hence, the importance of this formalization of evolutionary dynamics lies in its remarkable generality and simplicity. It is applicable to both asexual and sexual populations, to scenarios involving one or multiple alleles at different loci, and it can account for various factors, such as epistasis, group selection, and multi-level selection [26, 27].

A wide literature has analyzed the Price equation from both mathematical and philosophical perspectives [28, 20, 29, 30, 25, 31, 32], demonstrating its usefulness [33, 34] and its limitations [35, 36]. A comprehensive derivation of the final form of the equation is provided in the aforementioned literature; here, we aim to show how the equation is constructed and which quantities it explains.

While Fisher's model of selection is commonly used in population genetics, the Price equation is better suited to our model since we are not considering sexually reproducing organisms with the assumption of random mating. In our work, we have used this formalism as a framework into which we input data extracted from the model to observe whether changes in phenotypic traits, and fluctuations between parents' and offspring's phenotypic values, can be explained in terms of selective pressure and their relative importance in fitness. Below, we present one of the most commonly used forms of the equation:

$$\Delta\bar{\phi} = \frac{1}{\bar{W}}[cov(w,\phi) + E(w\bar{\delta})] \tag{2}$$

The term $\Delta\bar{\phi}$ on the left side represents the change in the mean phenotypic value. The first term on the right side of the equation, $\mathrm{cov}(w,\phi)$, measures how the fitness associated with the phenotype under observation changes over time. This covariance term is quite general, capturing changes caused by both selection and drift. Importantly, it measures a statistical association, reflecting changes directly attributed to differences in the contribution of the ancestral population to the size of the descendant population; that is, it quantifies the strength of the selective effect on that trait.

The second term, $E(w\bar{\delta})$, captures the expected differences in the trait values between parents and offspring, scaled by fitness. This term represents changes in the resemblance between ancestors and descendants, which can result from various events both within and outside individuals, such as variations in single genes or epigenetic effects (the latter are not included in our model).

The index i can refer to any single trait (allele, genotype, phenotype, etc.) or any individual identification belonging to a particular member of the population. Thus, in the present work, if $\phi_i = x$ represents the trait value of individual lineage i in the parental population, then $\phi_i' = x$ will represent the trait value in the descendant generation if it remains unchanged, or $\phi_i' = x + n$ if a mutation has occurred [37].

The linkage between changes in phenotype and subsequent fitness changes is expected in this model, as it reflects how we have programmed the cells to be-

have; nevertheless, valuable insights can still be gained. To perform an analysis of the model using the Price equation, we need to divide the total population into two sub-populations (ancestors and descendants) based on an arbitrarily chosen time interval; see Fig. 2. It is worth noting that this procedure may not capture all possible variations between individuals within and outside of this time interval. One can choose different time intervals, either longer or shorter, to examine evolutionary differences between the two sets of populations[3].

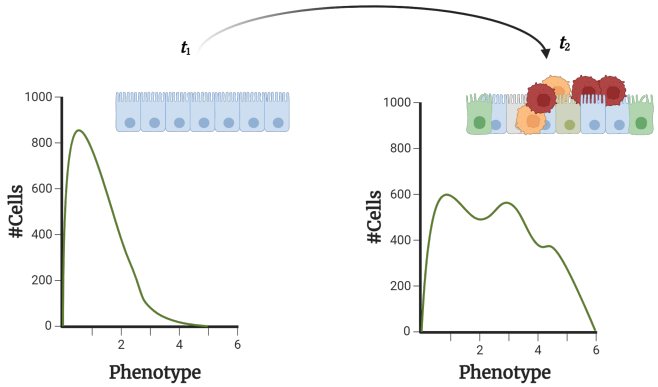


Figure 2: Here is depicted a possible change in the distributions of ancestor-descendant divided into two sub-populations. The individuals at t_1 are the *ancestors*, while the ones at t_2 are the *descendants*; different colors and shapes represent different phenotypes. Image built with biorender.com.

2.1.3. Genotype-Phenotype mapping

In classical population genetics, the fitness of an individual, w, refers to its reproductive success, the contribution of a given phenotype to the next generation, assuming a complete correspondence between genotypic and phenotypic traits regarding their contribution to the fitness change [38]. The genotype-phenotype mapping can be a complex issue, often represented as a multi-layered graph where various genotypes form the bottom layer, with arrows pointing to one or more phenotypes in the layer above. This complexity arises because one phenotype can be formed by different genotypes, and one genotype can give rise to more than one phenotype, indicating a many-to-many relationship with additional parameters involved [39, 40].

However, in our work the genotype-phenotype map is clearly defined: mutations in both copies of *APC* yield the *pre-adenoma* phenotype; adding a mutation in at least one copy of *KRAS* to the *pre-adenoma* phenotype results in the *adenoma* phenotype; if both copies of *TP53* are mutated, the *adenoma*

[3]See the additional material.

phenotype becomes *tumoral*; see Fig. 3. As a result, the mitosis time only changes in two cases: the mutation of one copy of *KRAS* and the emergence of the *tumoral* phenotype. In these cases, the mean mitosis time decreases from 24 hours for *physiological* and *pre-adenoma* cells to 12 hours for *adenoma* cells and 10 hours for *tumoral* cells. Following the aforementioned classical definition, we can obtain the proper fitness values by tracking the number of descendants per genotype/phenotype value, hence covering the whole genotype-phenotype space[4] [41].

Once we have the set of fitness values, we can associate them to the correspondent frequency plugging them into the recursive equation to track the allele frequency. Here is an example for $\Delta p = \Delta APC_{00}$.

$$\Delta APC_{00} = \frac{APC_{00}(1 - APC_{00})(w_{apc_{00}} - w_{apc_{11}})}{\bar{w}} \tag{3}$$

Unlike traditional population genetics, which primarily focuses on measuring changes in allele frequencies within a population, the approach using the Price equation is more general, encompassing everything from changes in allele frequencies to the characteristics of the entire organism within a population. In our crypt-world model, the phenotypic trait we aim to track across generations is the typology, which is tagged as *physiological*, *pre-adenoma*, *adenoma*, or *tumoral*; see Fig. 3. The reason behind this choice is the fact that the allele-frequency model is for one gene at a time, hence screening off the possible joint contribution of multiple genes to the fitness of the different typologies.

In genetic analysis, two types of phenotypic values can be assigned: quantitative and qualitative. A quantitative, or metric, trait is a characteristic of an individual or group that corresponds to a measurable value, such as height or weight. These traits, influenced by multiple genes, have been studied since the pioneering work of Fisher and Pearson, who highlighted the significant contribution of multiple genes to the final makeup of an adult individual [42].

However, some phenotypes exhibit discrete, qualitative characteristics, such as specific cellular typologies. To connect typology, a qualitative aspect, to a quantitative one, we have chosen to quantify them by summing the values of various gene lists. This is allowed by the structure of the genotype-phenotype map, which links the genetic values as the cause of the phenotypic values; consequently, the phenotypic value ϕ is calculated as the following sum:

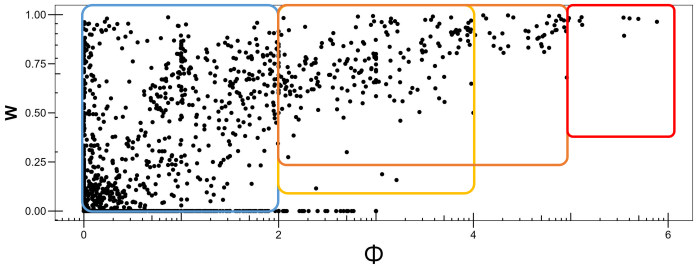
$$\phi = (\sum_{0}^{2} APC) + (\sum_{0}^{2} KRAS) + (\sum_{0}^{2} TP53)$$

The results span from $\phi = 0$, representing the basic *physiological* trait value where all genes are wild-type, to $\phi = 6$, indicating the *tumoral* trait value with

[4] Again, see additional material for more details.

APC	KRAS	P53	
00	00	00	Physiological
00	00	01	
00	00	11	
00	01	00	
00	01	01	
00	01	11	
00	11	00	
00	11	01	
00	11	11	
01	00	00	
01	00	01	
01	00	11	
01	01	00	
01	01	01	
01	01	11	
01	11	00	
01	11	01	
01	11	11	
11	00	00	Pre-adenoma
11	00	01	
11	00	11	
11	01	00	Adenoma
11	01	01	
11	01	11	Tumoral
11	11	00	
11	11	01	
11	11	11	

[a]



[b]

Figure 3: In [a] is depicted the genotype-phenotype map as a table of values. Notice that here are emphasized, with different colors, only the minimum values that trigger the change in cell typologies. However, other combinations present in the chart are functionally different from the physiological one. For instance, in the fourth line the genotype [$APC = 00$, $Kras = 01$, $TP53 = 00$], gives to the bearing cell a reduced mitosis time, because *Kras* has one mutated copy. Also there are more physiological genotypes other than the [$APC = 00$, $Kras = 00$, $TP53 = 00$]. In the second line we have a genotype, [$APC = 00$, $Kras = 00$, $TP53 = 01$], still in physiological state, since one mutation in *TP53* is not enough to trigger different actions. The structure is clearly visible in the plot [b] where is reported the distribution of fitness W in function of the phenotype value ϕ. From the left: the blue rectangle is over the physiological values [0 to 2]; the yellow one is on the *pre-adenoma* values [2 to 4]; the orange one is on *adenoma* values [2 to 5]; finally, the red one is on the *tumoral* values [5 to 6].

all the genes being mutated. Between these two extreme values lie various possible combinations, some neutral and others corresponding to neoplastic variants, again refer to Fig.(3).

3. Results

In this section, we present the results obtained by setting the mitosis time of neoplastic cells and then removing it. The first condition ensures a differ-

ence in mitosis capacity, while the second condition only leaves the difference in movement as the discriminant feature between cell phenotypes. To recall the aforementioned specifics of the model: *physiological* cells have an average mitosis time of 24 hours; *pre-adenoma* cells keep the same average mitosis time; *adenoma* cells have a decreased average mitosis time of 12 hours; *tumoral* cells have a decreased mitosis time of 10 hours. Once these differences are removed, all the cells have a basic mean mitosis time of 24 hours; nonetheless, they move differently in relation to typology. *Physiological* cells move straight ahead; *pre-adenoma* and *adenoma* cells deviate by a certain degree either to the left or to the right; *tumoral* cells are static but can proliferate in all directions; see Fig. 1.

3.1. Results with Classic Selection Model

Our goal is not only to observe changes in allele frequencies as mutation probability increases, since such changes are expected by definition, but also to gain insights into which genes are more susceptible to selective pressure. To measure selective pressure, we have plotted the variation, Δp, as a function of the wild-type homozygote allele frequency, p, itself. This approach allows us to observe whether the wild-type allele frequency of the three genes correspond to a positive, negative, or null variation. For a visual representation of these dynamics, refer to (Fig. 4).

We analyzed allele frequency dynamics of the three driver genes (*APC*, *KRAS*, and *TP53*) under increasing mutation rates ($\mu = 10^{-9}$, 10^{-6}, 10^{-4}, 10^{-2}). The immediate difference an observer can spot is the complete range of *KRAS*, from $p = 0$ to $p = 1$, in the scenario with fixed mitosis time. This is due to the fact that *KRAS* is the only gene carrying effects on mitosis even when mutated in one copy. At physiological mutation rates ($\mu = 10^{-9}$), all three genes exhibited positive Δp across most values of p, indicating strong selection favoring wild-type alleles, particularly in *APC* and *TP53*.

At $\mu = 10^{-6}$, we observed a tail for $p \leq 0.5$ lowering the positive selection mostly for *APC* and *TP53*, while for $p > 0.5$ the shape of the curves remained unchanged. This is the first scenario where the dependence of gene selection on differences in the decreased mitosis time is clear. In Fig. 4 (b and d), in fact, all three genes showed roughly the same trend for $p > 0.5$, while in 4 (c), *APC* started to show less positive selection near $p = 0.5$.

As mutation rates increased to $\mu = 10^{-4}$, the curves became noisier, reflecting a mutation-selection balance, with *KRAS* and *APC* showing negative selection for $p \leq 0.4$. At this mutation probability, even in the simulations without fixed mitosis time, we observed lower positive selection; moreover, *APC* was under negative selection for $p \leq 0.4$ as well as in the standard scenarios.

At $\mu = 10^{-2}$, the curves were more symmetric around $\Delta p = 0$, showing that the equilibrium points for the three genes were all at $p \leq 0.5$; below that frequency they were under strong negative selection. Here a notable shift was observed: *TP53* and *KRAS* displayed negative Δp at low p, suggesting a reversal in selection direction and potential positive selection for mutant alleles. In

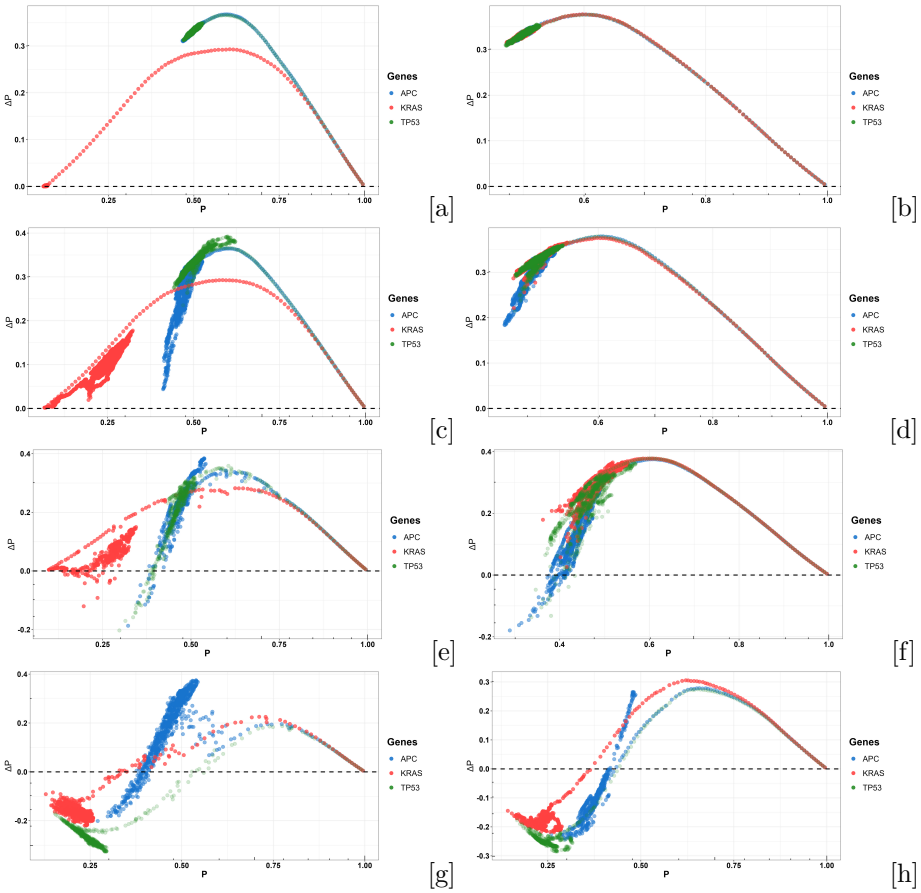


Figure 4: This series of plots shows the four chosen scenarios ($10^{-9}, 10^{-6}, 10^{-4}, 10^{-2}$, from the top to the bottom), divided by the condition of fixed mitosis time (left with fixed mitosis time, right without).

contrast, *APC* maintained strong purifying selection across all regimes, highlighting its greater resistance to mutation-driven allele frequency shifts.

First, these findings demonstrate how increasing mutation pressure can erode purifying selection and even invert selective dynamics in key driver genes. While environmental factors are active and the mutation probability is close to the physiological one ($\mu = 10^{-6}$), neoplastic allele variants cannot overtake wild-type alleles while frequency is below 0.5.

Moreover, results suggest that selection against wild-type variants of genes is not the same at different μ values. For instance, in the fixed-time setup at $\mu = 10^{-6,-4}$, *APC* and *TP53* have greater negative selection than *KRAS*, showing that non-differentiated cells with damaged DNA are favored over physiological ones. At the same time, somewhat paradoxically, *APC* at $\mu = 10^{-2}$ presents strong positive selection in a small interval near $p = 0.5$.

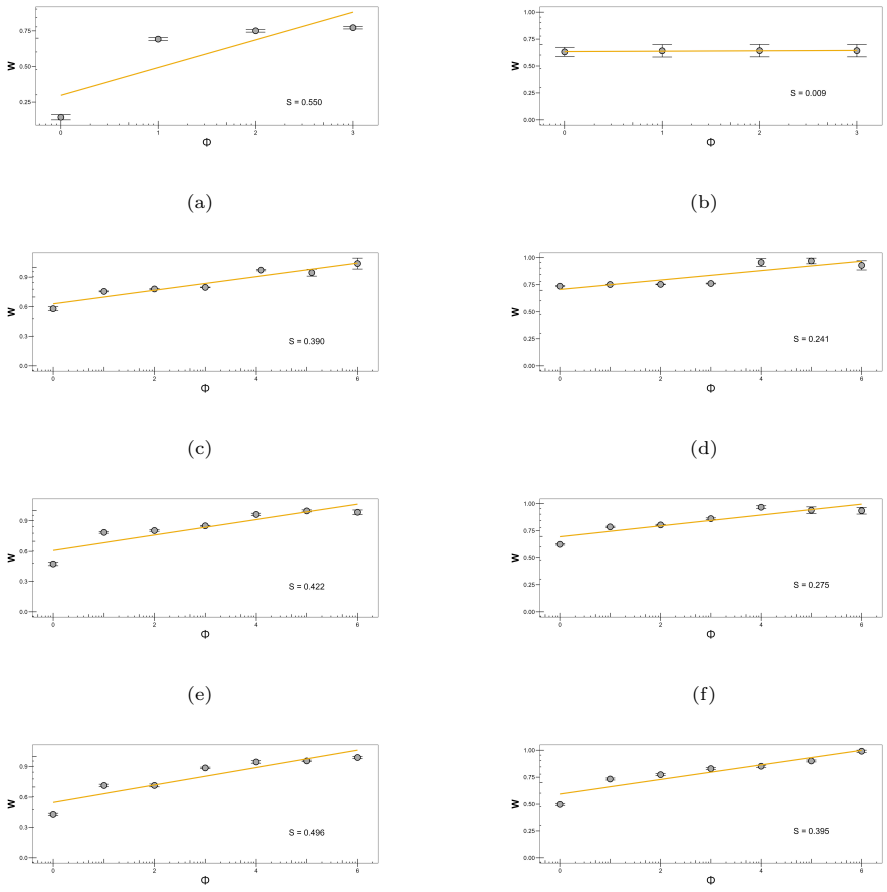
3.2. Results with Price equation information

In this section we present the results on the relation between phenotype values and fitness values, informed by the Price equation, see (Fig. 5 and 6).

First, we focus on the plots presenting the linear relationship between trait values, ϕ, and fitness, W [20]. In a previous figure, Fig. 3 [b], we reported the phenotypes corresponding to certain trait values highlighted with different colors. Applying the same criterion for the plots under consideration, we observe that increasing μ corresponds to a rise of different phenotypes and their associated fitness, following a linear trend similar in both fixed and non-fixed mitosis time scenarios.

The comparison between the two setups at $\mu = 10^{-9}$ shows a dramatic difference, suggesting that variable reproductive timing unlocks adaptive potential that is hidden when all individuals reproduce with the same average mitosis time. The fixed-time scenario highlights a strong positive linear relationship between phenotype and fitness, with W increasing from approximately 0.15 to 0.80 as ϕ increases from 0 to 3. This is a clear adaptive response guided by the advantage of *Kras* heterozygotes. The flat relationship in the non-fixed-time scenario indicates essentially no fitness response to selection. Despite variation in phenotype (ϕ), fitness (W) remains constant around 0.65 across all phenotypic values. All the phenotypic variation is due to the initial probability of stem cells having a homozygote or heterozygote genotype.

Somewhat counterintuitively, things begin to change when we increase mutation probability. At $\mu = 10^{-6}$, all phenotypes emerge showing a linear increasing trend with an evident leap from $W = 0.8$ at $\phi = 3$ to $W = 0.95$ at $\phi = 4$, with the maximum fitness value reached by the tumoral cells at $\phi = 6$. So far the expected behavior has occurred; however, if we focus on S, we realize that it is considerably lower than in the physiological scenario. This is due to the features of the physiological scenario where the only cause of different fitness outcomes is the presence of *Kras* heterozygotes, which enable bearing cells with $\phi = 1, 2, 3$ to enter mitosis more rapidly. Therefore, it is a difference related only to reproductive success, whereas at $\mu = 10^{-6}$, even though there are more neoplastic



(a) (b) (c) (d) (e) (f) (g) (h)

Figure 5: This series of plots shows the four chosen scenarios ($10^{-9}, 10^{-6}, 10^{-4}, 10^{-2}$, from the top to the bottom), divided by the condition of fixed mitosis time (left with fixed mitosis time, right without). In the x-axis is reported the mean phenotypic value ϕ, in the y-axis is reported the mean fitness W. In all the plots is reported the selective gradient S, used as an intuitive measure of the selective strength.

phenotypes with high fitness values, environmental factors start to play a significant role in shaping population traits. Environmental factors are at the core of crypt homeostasis when neoplastic variants are present. While in physiological conditions, cells capable of undergoing more mitoses without endangering the entire crypt architecture are favored by selection (thus the high S), in scenarios with triggered environmental conditions, highly neoplastic cells are favored, though not as much as cells with different phenotypes in a physiological crypt.

While in the fixed-time scenario there is a composition of three main trends ($\phi = 0$, $\phi = 1, 2, 3$, $\phi = 4, 5, 6$), in the non-fixed one two moments are observable ($\phi = 0, 1, 2, 3$, $\phi = 4, 5, 6$), engendered by the difference in movement of cells with $\phi = 4, 5, 6$. Broadening the meaning of environmental factors to include crypt architecture, we can explain the trend in the non-fixed time scenario as well. It is documented that neoplastic cells undergoing epithelial-mesenchymal transition (EMT) acquire motility and evolutionary plasticity in relation to the environment [43, 44], thus enhancing neoplastic cell invasion and proliferation more than the physiological ones. This is the same behavior programmed in cells with phenotypes starting from *pre-adenoma*, which move and reproduce in different directions regardless of their surroundings, i.e., other cells, thus producing a different mitosis pattern as shown in Fig. 5d.

The remaining scenarios at $\mu = 10^{-4}$ and $\mu = 10^{-2}$ do not present particular differences in the nature of the trend. There is an expected increase of S in both fixed-time setups, but not as high as in the physiological one, and the three aforementioned moments of the trend are almost equal; the only difference is at $\mu = 10^{-2}$, where $\phi = 3$ is also close to the values of the second moment ($\phi = 4, 5, 6$). The non-fixed scenarios highlight even more the role of differential movement in accelerating mitosis events, with *tumoral* cells always reaching the highest values even while not moving at all, yet displacing descendants in every direction.

Another set of relevant results provides useful information directly from the Price equation, see Fig. 6.

The four terms are the covariance between phenotype and fitness $\mathrm{cov}(\phi, w)$, which is the first term on the right-hand side of the Price equation; the transmission bias, $\bar{\delta}$ – that is to say, the difference between parents' and offspring's mean trait values – which is the second term; the left-hand term of the Price equation $\Delta\bar{\phi}$ – in other words, the main information we want to obtain from the Price formalism; and the variance of the trait value, which is a particularly useful element for understanding how traits vary across different μ values.

We are now considering the $\mathrm{cov}(\phi, w)$ term, which is the element of the equation that captures natural selection on the parental generation. Following the model features, selective pressure is expected to be an increasing quantity from $\mu = 10^{-9}$ to $\mu = 10^{-2}$, and is expected to be lower or null in the setup with non-fixed mitosis time. Curiously, in the fixed-time scenario, the covariance at $\mu = 10^{-9}$ is larger, near 0.10, than at $\mu = 10^{-6,-4}$, near 0.05; nevertheless, at $\mu = 10^{-2}$ it rises to 0.20. This trend confirms that, even within the chosen short time step of 24 hours, the presence of environmental constraints at $\mu = 10^{-6,-4}$ hampers the fitness difference potential of higher-value phenotypes.

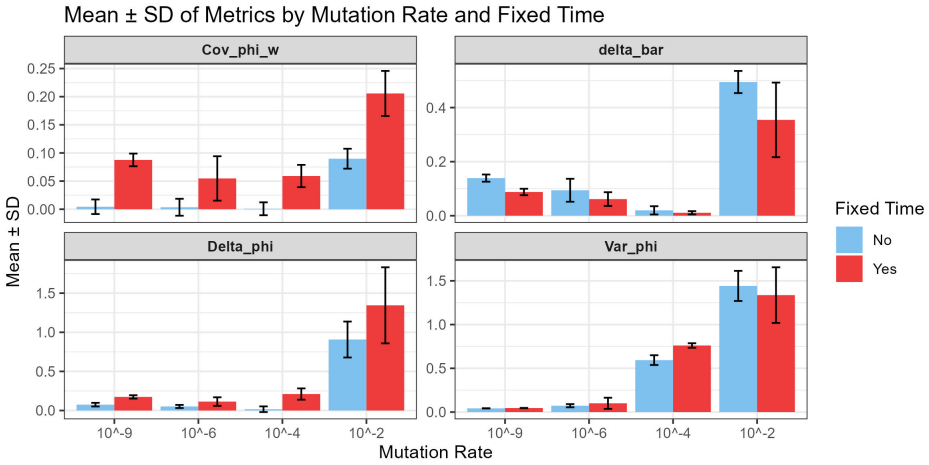


Figure 6: Relevant measures for all the simulated scenarios across multiple runs, categorized by scenarios with fixed mitosis time, and without fixed mitosis time (fixed time: yes in red, no in light blue). From the top left: covariance, $cov(\phi, w)$; difference in the mean trait value between parents and offspring, $\bar{\delta}$; result of the Price equation, that is change in the mean trait for the correspondent time interval, $\Delta\bar{\phi}$; variance of the population trait value $var(\phi)$.

It also confirms that the only element of change in the physiological scenario is the enhanced mitosis capacity due to *KRAS* mutation, hence a difference in fitness which is the hallmark of natural selection alone, precisely captured by the $\text{cov}(\phi, w)$ term. The non-fixed time setup behaves exactly as expected since the covariance is very low, null, and even negative, reaching almost 0.10 only at $\mu = 10^{-2}$. This is probably occurring because tumoral cells have so many degrees of freedom, such as direction of reproduction, that the number of mitoses is not comparable to a significant difference in fitness.

The transmission bias term ($\bar{\delta}$) accounts for mechanisms other than selection and drift. Arguably, this is the most peculiar result of the model, since scenarios with fixed-time mitosis show lower values of difference between the two generations than the non-fixed-time ones. A reasonable explanation for the higher values of $\bar{\delta}$ in non-fixed-time scenarios is that selection maintains traits ensuring higher fitness in fixed-time scenarios. Cells with a beneficial mutation transmit their phenotype to the offspring, and the offspring will proliferate keeping the same phenotype, hence reducing the difference between generations after the first one. In non-fixed time scenarios, all phenotypes have the same average mitosis time, thus transmitting differences in movement freedom. However, the difference in movement is not directly linked to a selective advantage, as shown in Fig. 5, hence more variability due to random mutation can occur, increasing the difference between offspring and parents.

The other interesting phenomenon is the decrease of the term as the mutation probability increases. We can explain this by pointing to the presence of environmental factors targeting neoplastic cells and killing them. Both hypoxic

conditions and immune-system cells make the target cellular population size drop from the peak, thus resulting in an abrupt change of phenotype values. Nevertheless, what is left is the number of cells of the same lineage with a lower phenotypic value. Suppose one cell with lineage n divides and, of the two daughter cells, one mutates and the other does not. What happens after a certain period is that the two groups have the same lineage but different phenotypes, hence one group will trigger environmental conditions, eventually dying, while the other will survive, lowering $\bar{\delta}$ in the chosen time frame.

The change in the mean trait, $\Delta\bar{\phi}$, reflects what we derived for the two components of the Price equation, revealing that the overall change, from one generation to another, is greater when a substantial difference in fitness values is present. Variance too essentially follows the same trend, showing greater phenotypic heterogeneity as the mutation probability increases [45].

4. Discussion

A fundamental challenge in biological modeling is the attribution of biological meaning to simple mathematical formulations, which are used to isolate mechanisms and disentangle causal relations. The mathematical models employed to examine the evolutionary dynamics of the simulated crypt are indeed rather simple; nevertheless, they were able to capture several relevant regularities consistent with evolutionary theory. This "law-seeker" approach does not propose a general model for real tumor evolution, as it lacks several biological and ecological features; rather, it aims to offer a general description of the model itself as an abstraction of the real system. The ABM, together with its analysis through evolutionary biology formalisms, has been conceived as a theoretical tool to isolate behaviors and trace back their causes through fine-grained control over both genetic and environmental variables [46, 47].

However, precisely because of its versatility, the model can help elucidate several theoretical questions, such as ecological interactions or cellular motility as key elements of tumor progression. For instance, it is debated whether predator–prey interaction represents a suitable explanatory framework for the interaction between tumoral cells (prey) and immune system cells (predators) [48]. Analyzing the dynamics between neoplastic cells and immune-system cells in our model could provide information to confirm or reject the predator–prey model.

Invasiveness and metastasis are challenges of paramount importance for cancer therapy and are tightly linked to EMT and the stem-like properties of tumoral cells, which begin to ignore signals from physiological tissues and neighboring cells. Various in silico models have shown how different movement and space-distribution patterns align with experimental findings on metastatic processes [49]. Calibrating the movement degrees and proliferation directions to those of other models would lead to unified frameworks as the basis for more sophisticated in silico systems, such as digital twins [50].

Furthermore, as long as certain features of the model can be replicated for the study of different cancers, the model itself can contribute to the theoretical

understanding of tumor progression in other tissues and organs. For instance, a statistical association between mutations in TP53 and KRAS has been observed in ovarian epithelial cancer [51] and in lung adenocarcinomas [52]. In these cancer types, the activity of both driver genes can be roughly compared to those represented in our model – namely, increased proliferation, altered cell adhesion, and DNA damage leading to a higher mutation rate. Therefore, after adapting the crypt structure to reflect the characteristics of these tissues, it may be possible to investigate the roles of these genes in different biological environments.

These efforts to build robust, multi-scale models with predictive power naturally complement the possibility of identifying general laws in biology and ecology – defined as statements about invariant properties under given conditions – which appears plausible if the boundaries of the explanatory system are clearly defined [53]. Thus, once some small-world laws derived from real-world data are established, it would be possible to construct theories about the underlying mechanisms governing their behavior. These mechanisms would, in principle, be able to describe, explain, and predict features of the model, even when novel characteristics of the real world are incorporated. For example, if an energy consumption rate linked to a different set of genes were introduced, the same framework developed to study cellular movement should be suitable for explaining the new phenomenon. In our case, that same framework would be the Price equation, accounting for selection and other evolutionary mechanisms.

Nevertheless, precisely because we are operating within a highly simplified computational model, some issues remain unresolved. Why, for instance, do both setups produce the same trends with only a small difference detected by S? Does that mean that different movements, arising from a given genotype, are the real cause of the difference in the number of mitoses, whereas the fixed average mitosis time exerts only a mild influence? Is the fact that wild-type alleles continue to experience positive selection an artifact of the selection model, or do homeostatic mechanisms remain efficient even in such a simplified landscape?

These discrepancies could be related to the assumed linearity of the phenomena we are observing [54, 55, 56]. In real systems, such as the colonic crypt, a high mutation probability results from a considerable number of factors that strongly deregulate cellular function at all levels, from the genetic to the population scale. These factors cannot simply be summarized and represented by a linear trend, as they influence one another in complex ways, complicating the causal network. Even in an oversimplified model such as the one examined here, changes in individual and group-specific traits lead to population effects that, in turn, affect individual traits – specifically, hypoxic threshold cause all the cells in that spot to die even if they are *physiological*. Otherwise, the phenotypes and their relative fitness would increase in a simple linear manner, explainable by S alone.

However, as observed by Rice [54], increasing the polynomial degree of the regression equation can indeed improve data fitting, though at the cost of the biological meaningfulness of the values.

The linear assumption underpinning the Price equation provides useful, al-

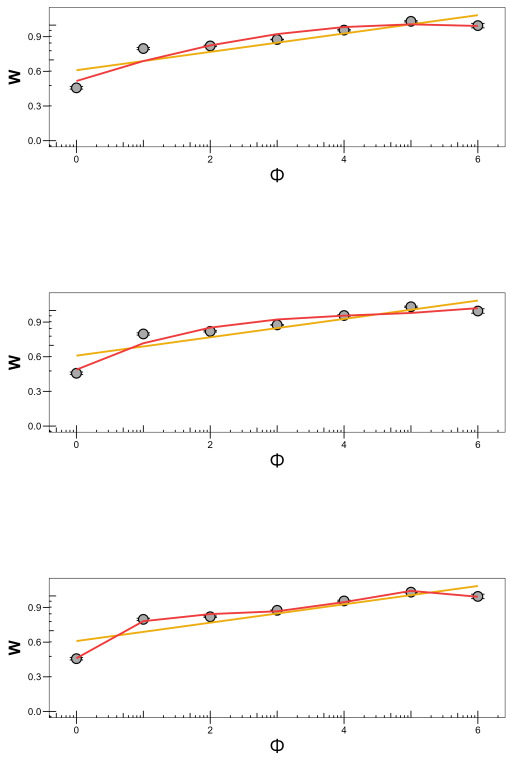


Figure 7: Three plots showing fit lines for increasing polynomial degrees in red, and the linear fit in yellow. From the top: x^2, x^3, x^4.

though incomplete, information about the relationship between phenotype and reproductive success, survival, or any other outcome linked to the perpetuation of a trait across generations. In fact, the first-term coefficient in the linear trend precisely represents the strength of selection on the trait, while the coefficients of higher-order polynomials lose any biological interpretability beyond mere description. In Fig. 7, higher polynomials indicate that something is missing from the linear model; however, their coefficients are not related to any biological mechanism (selection) or phenomenon (average fitness). Thus, by balancing simplifying assumptions and biological realism, this modeling approach contributes to the broader challenge for generalizable laws governing complex biological systems.

5. Conclusions

In this study, we performed several simulations with the aim of gaining valuable insights on the existence of natural selection mechanisms and how these mechanisms drive the behavior of both individual cells and cell populations. Interesting conclusions can be drawn from the results obtained when predetermined mitosis time was not used, i.e. different degrees of freedom in movement lead to different fitness values. It is biologically reasonable to assume that for physiological behavior at the individual and population level, tissue architecture (movement allowed in one direction only) is a limit for the individual cellular activity. Hence, when individual mutations deregulate local sub-populations, macro-level boundaries are altered or are turned ineffective in the limitation of individual behavior (allowing movement in all directions), thus leading to abnormal and dangerous phenomena such as uncontrolled mitosis. A further conclusion can be drawn from the results indicating that environmental conditions play a relevant role in the interplay of different neoplastic phenotypes. These interactions are not present in the simulations of the crypt, in which the cells of the immune system and the hypoxic thresholds are inactive. From that we can reasonably deduce that, when considering an evolutionary perspective of tumor progression, we must consider environmental factors not only as background conditions, but as true causal (a component of the selective mechanism) factors alike the activation of oncogenes and the silencing of tumor suppressor genes by mutational events.

We believe that these results place the model in the third category of Levins's classification [57], which is to sacrifice precision for realism and generality. A certain degree of realism is implicit in the model because, although it is based on simplified assumptions, it relies on experimental and clinical data. Therefore, as long as the importance of driver genes in tumor development is confirmed by the majority of clinical samples, a small part of reality is captured by the simulated world. Generality refers to the importance of local behavior in generating emergent organizational properties that counteract shaping the evolution of the whole system. In the simulated world, the local behavior is characterized by the movement and the reproduction direction of the agents, which are almost universally shared properties among different living beings [58]. These

local features in turn triggers population thresholds which cause either death or opportunity for the individuals belonging to different cellular populations [59, 60].

Our future research efforts aim to formally define the causal evolutionary structure underlying the model and to clearly dissect the selective agents to gain insights into their weight in the progression of CRC. Moreover, these results will also be useful to shed light on the concept of top-down causal relationships or higher level constraints and how they can be formalized using the language of causal inference [61, 56].

6. Appendix

Datasets and computational model are available at the following links:

Google documents (necessary for the size of the data), `https://drive.google.com/drive/folders/1iLLbqHcUKkQKO_qRvBZ8I92IGspZdF5_?usp=sharing`

Github, `https://github.com/MLedda7/CRC-data-models`

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